THE PROBLEM

Predict a continuous target from 15 anonymized features

METRIC

R^2

Higher is better

THE JOURNEY

Score Progression

-0.46	→	-0.093	→	0.04875	→	0.05122
Naive		OLS		Ridge		Tuned + Feature Drop

DISCOVERY

What Went Wrong

$R^2 = -0.46$

Naive Linear Regression

Worse than predicting the mean

- Target = weak linear signal + ~3% corrupted outliers (±40,000)
- Polynomial, product, ratio, tree-based hypotheses all rejected
- Outliers are irreducible noise — unpredictable by any model

INSIGHT

Why R^2 Is Dominated by Outliers

NORMAL ROW

`target ≈ 30`

Contribution: ~900

to denominator variance

OUTLIER ROW

`target ≈ 40,000`

Contribution: ~1,600,000,000

1.8 MILLION × one normal row

~3% of rows control ~95%+ of the metric

THE TRAP

Trimming Outliers

TRIMMED CV R^2

0.95

Looks beautiful!

FULL-DATA R^2

0.005

Reality.

“You optimized the 5% of variance the metric barely cares about”

SOLUTION

The Final Strategy

1. **Full Dataset**
   Train on ALL data including outliers — because R^2 rewards it

2. **Ridge Regression**
   Heavy regularization ($\alpha = 4000$) resists outlier leverage

3. **Stability Validated**
   40-seed test: mean +0.041, never negative

4. **Feature Selection**
   Dropped 3 noise features (x11, x12, x13)

RESULT

0.051

vs 0.005
from trimming

COMPARISON

Trimmed vs Full Data

	Trimmed (Trap)	Full Data (Correct)
Optimizes	Clean core (5% of variance)	Full distribution (95% from outliers)
CV R^2	0.95 (misleading)	~0.04 (honest)
Kaggle R^2	**0.005** ❌	**0.051** ✅
Why	Distribution mismatch	Matches the real metric

PROCESS

What Was Done

- ✓ Explored before modeling
- ✓ Formed and tested hypotheses (didn't assume)
- ✓ Caught the CV-vs-reality trap with honest holdout
- ✓ Used limited submissions strategically
- ✓ Reached realistic ceiling with simple, defensible model

KEY TAKEAWAY

- Trimming is only safe when outliers are absent from test data
- OR your metric is robust to them (e.g., MAE, not R^2)
- When metric is dominated by noise: maximize stability, not fit